EX 10 IMAGE GENERATION USING GENERATIVE ADVERSARIAL NETWORK (GAN)

DATE:

Problem Statement:

Train a Generative Adversarial Network (GAN) to generate new images from a dataset. Evaluate the quality of generated images using both visual inspection and a quantitative metric such as Inception Score (IS) or Fréchet Inception Distance (FID).

Suggested Dataset: LSUN Dataset

Objectives:

1. Understand the architecture of GANs and their adversarial training process. 2. Use StyleGAN2-ADA, a state-of-the-art GAN, for high-quality image generation. 3. Evaluate generated images using IS and FID metrics.
4. Compare generated outputs with real samples and assess realism.

Scope:

GANs are powerful generative models used in image synthesis, super-resolution, style transfer, and more. This experiment familiarizes students with modern GAN architectures and helps develop understanding in both visual and metric-based evaluation of generative quality.

Tools and Libraries Used:

1. Python 3.x
2. PyTorch
3. torchvision
4. StyleGAN2-ADA
5. torchmetrics
6. FID & Inception Score libraries
7. CelebA Dataset

Implementation Steps:

Step 1: Clone StyleGAN2-ADA Repository

```
REPO = 'stylegan2-ada-pytorch'
REPO_URL = 'https://github.com/NVlabs/stylegan2-ada-pytorch.git'

def clone_repo():
 if not os.path.exists(REPO):
 subprocess.run(['git', 'clone', REPO_URL])
```



```
 sys.path.append(os.path.abspath(REPO))
```

Step 2: Download Pretrained Generator

```
MODEL_URL = 'https://nvlabs-fi-cdn.nvidia.com/stylegan2-ada
pytorch/pretrained/ffhq.pkl'
MODEL_PATH = 'ffhq.pkl'

def download_model():
 if not os.path.exists(MODEL_PATH):
```

```
 import urllib.request
 urllib.request.urlretrieve(MODEL_URL, MODEL_PATH)
```

Step 3: Generate Synthetic Images

```
def generate_images():
 import legacy
 import dnnlib
 G = legacy.load_network_pkl(open(MODEL_PATH, 'rb'))['G_ema'].to(device)
os.makedirs('generated_images', exist_ok=True)
 imgs = []
 for i in range(1): # NUM_IMAGES = 1
 z = torch.randn([1, G.z_dim], device=device)
 label = torch.zeros([1, G.c_dim], device=device)
 img = (G(z, label, truncation_psi=0.5, noise_mode='const') + 1) * 0.5
save_image(img, f'generated_images/fake_{i}.png')
 imgs.append(img.cpu())
 return torch.cat(imgs)
```

Step 4: Calculate Inception Score

```
from torchmetrics.image.inception import InceptionScore

def calculate_inception_score(imgs):
 loader = DataLoader(TensorDataset((imgs * 255).clamp(0, 255).to(torch.uint8)),
batch_size=32)
 is_metric = InceptionScore()
 for batch in loader:
 is_metric.update(batch[0])
 mean, std = is_metric.compute()
 print(f"Inception Score: {mean:.2f} ± {std:.2f}")
```

Step 5: Prepare Real Images from CelebA

```
def prepare_real_images():
 dataset = dsets.CelebA(root='./data', split='train',
 transform=transforms.Compose([
 transforms.Resize(512),
 transforms.CenterCrop(512),
 transforms.ToTensor()
 ]),
```



```
 download=True)
 loader = DataLoader(dataset, batch_size=1, shuffle=True)
 os.makedirs('real_images', exist_ok=True)
 for i, (img, _) in enumerate(loader):
 if i >= 1: break
 save_image(img, f"real_images/real_{i}.png")
```

Step 6: Compute FID Between Real and Generated Images

```
def calculate_fid():
 subprocess.run(["python", "-m", "pytorch_fid", "real_images", "generated_images"])
```

Step 7: Run Entire Workflow

```
def main():
```

```
clone_repo()
download_model()
imgs = generate_images()
calculate_inception_score(imgs)
prepare_real_images()
calculate_fid()

if __name__ == "__main__":
main()
```

Conclusion:

This experiment highlights the capabilities of advanced GANs like StyleGAN2 in generating photorealistic images. The use of Inception Score and FID provides quantitative support to visual assessments. GANs remain a cornerstone in generative deep learning research, with wide applications in media, design, and beyond.

SAMPLE EXPECTED OUTPUT

Experiment 10: Image Generation using GAN

Pretrained StyleGAN2-ADA image synthesis and metric evaluation

Illustrative only. The experiment was not executed. All example values, curves and text below are invented to show the expected output format; they are not measured results.

Sample console / output format

```
Expected saved files:
generated_images/fake_0.png
real_images/real_0.png

Metric output format after valid evaluation:
Inception Score: <mean> +/- <standard deviation>
FID: <computed value>
```

Output	Expected form
Generated sample	One 1024 x 1024 RGB image from the FFHQ model
Real comparison sample	One CelebA image resized/cropped to 512 x 512
Evaluation metrics	IS and FID on suitably sized image sets

How to read this output

The supplied code uses a pretrained FFHQ generator rather than training on LSUN. One image is insufficient for meaningful IS/FID; the metric step may fail. Numeric scores are intentionally omitted. A valid evaluation needs many samples and a consistent protocol. No actual GAN image is claimed here.